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Computer Engineering

DELIVERABLES OF THE FOURTH MODULE

Omnivest Project

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ABSTRACT

This document presents a consolidated public report of the Omnivest project, integrating the outcomes of Modules 1, 2, 3 and 4 covering Sprints 1 through 15. The report follows the formatting and narrative structure established, while unifying all technical, architectural, and business deliverables developed throughout the project lifecycle. The consolidated view demonstrates Omnivest's evolution from conceptual planning and data preparation to a secure full-stack platform, scalable cloud infrastructure, advanced predictive intelligence, and validated business strategy. Together, these modules establish Omnivest as a robust, AI-driven, and investment-ready financial analysis platform.

INTRODUCTION

The Omnivest project is an AI-driven financial analysis and decision-support platform designed to democratize access to intelligent investment insights. Developed through a structured, modular, and sprint-based methodology, the project progressed across three major development modules.

Module 1 established the conceptual, methodological, and data foundations of the platform. Module 2 transitioned the system into a secure and functional full-stack application with authentication, dashboards, and modular backend services. Module 3 focused on scalability, predictive intelligence, cloud deployment, and business validation. This consolidated report documents the objectives, activities, and key deliverables of all fifteen sprints, providing a unified and coherent overview of the complete development cycle.

CONSOLIDATED MODULE DELIVERABLES

Sprint 1 – Project Planning and Methodology Definition

Sprint 1 laid the strategic foundation of the Omnivest project. The team defined project scope, objectives, timeline, and key deliverables through stakeholder alignment meetings. Roles and responsibilities were assigned, agile methodologies were selected, and a high-level technology stack was discussed. This sprint resulted in a structured project charter and a shared roadmap guiding all subsequent development phases.

Sprint 2 – Requirements Engineering and User-Centered Design

Sprint 2 focused on aligning system objectives with user needs. Stakeholder interviews and workshops were conducted to gather functional and non-functional requirements, resulting in a formal requirements specification document. User personas and journey maps were created to model system interactions, and low-fidelity wireframes were designed using Figma. A preliminary system architecture draft was also produced, initiating early technical planning.

Sprint 3 – Data Acquisition and Preparation

Sprint 3 established the data foundation required for AI-driven functionality. Relevant financial datasets were identified, collected, cleaned, and normalized. The team handled missing values, outliers, and inconsistencies, and partitioned the dataset into training, validation, and test

subsets. Automated preprocessing scripts were developed to ensure reproducibility and scalability, ensuring data readiness for machine learning experimentation.

Sprint 4 – Model Evaluation and Selection

Sprint 4 focused on evaluating machine learning algorithms suitable for predictive analytics and financial optimization. The team conducted controlled experiments comparing interpretability, accuracy, and computational efficiency. Models such as linear regression, decision trees, and XGBoost were selected based on documented benchmarks and comparative analysis, establishing a clear rationale for implementation.

Sprint 5 – Initial AI Model Implementation

Sprint 5 marked the first implementation of Omnivest's intelligent core. Selected models were coded using Python and relevant machine learning libraries and integrated into a modular framework. Early prototypes were tested with both synthetic and real datasets, and unit testing was conducted to validate functionality. This sprint validated technical feasibility before further scaling.

Sprint 6 – Technical Architecture and Planning

Sprint 6 defined the architectural vision for the system's full-stack evolution. The frontend was planned using React and Tailwind CSS with modular components, while FastAPI was selected for backend services. Secure authentication using JWT was planned, and wireframes for dashboards and authentication flows were finalized. A detailed implementation plan and GitHub roadmap concluded the sprint.

Sprint 7 – User Interface Implementation

Sprint 7 translated design prototypes into a functional user interface. Dashboard, Portfolio, Transactions, and Settings pages were implemented with reusable components. Company filtering and data visualization were integrated using dummy JSON data, and Recharts was adopted for financial graphs. Responsiveness and accessibility were ensured through Tailwind CSS.

Sprint 8 – Authentication and Security Backend

Sprint 8 delivered a complete authentication system. The backend was implemented using FastAPI, SQLite, and SQLAlchemy, with password hashing via bcrypt. OAuth2-compliant JWT-based authentication enabled secure login, registration, password updates, and protected routes. Frontend integration ensured token validation and session control.

Sprint 9 – Backend Refactoring and Data Structuring

Sprint 9 focused on improving backend maintainability and scalability. Business logic was modularized into extractor, transformer, and writer components. JSON output structures were standardized, indexing company data by name to improve readability and downstream processing. These changes enhanced debuggability and long-term extensibility.

Sprint 10 – Testing and Stabilization

Sprint 10 emphasized system reliability and quality assurance. Unit tests were implemented for backend endpoints, JWT authentication flows were validated, and bugs related to file parsing and user sessions were resolved. This sprint ensured system stability and readiness for future integrations.

Sprint 11 – Cloud Architecture on AWS

Sprint 11 defined Omnivest's scalable cloud infrastructure on AWS. The architecture integrated Amazon Cognito for authentication, AWS WAF for security, CloudFront and S3 for static hosting, Lambda and EC2 for compute, and DocumentDB and RDS for data persistence. The design emphasized scalability, reliability, and production readiness.

Sprint 12 – Prediction Service and Model Refinement

Sprint 12 delivered a structured prediction service connecting financial data to the AI model. Input validation, dataset preparation, and model communication were implemented. The predictive model was refined through feature engineering, normalization, multicollinearity control, and automated model comparison using PyCaret, improving reliability and interpretability.

Sprint 13 – Prediction API and Data Extraction Overhaul

Sprint 13 implemented a FastAPI-based prediction API, enabling real-time financial forecasts via REST endpoints. Robust error handling and data cleaning logic were introduced. Additionally, the data extraction pipeline was reengineered, replacing regex-based parsing with a tokenization-based approach, significantly improving robustness and maintainability when processing financial PDFs.

Sprint 14 – Data Standardization and Model Documentation

Sprint 14 standardized the internal company data JSON schema, organizing financial metrics by year with consistent metadata and sector classification. Documentation was created detailing how the predictive model consumes the data, including lagged and year-over-year features, ensuring clarity, scalability, and future-proofing.

Sprint 15 – Business and Financial Planning

Sprint 15 transitioned Omnivest into a market-oriented product. The team developed a comprehensive business plan, including value proposition, market analysis, financial model, and funding strategy. A 12-month financial projection estimated operational costs and evaluated funding scenarios, aligning technical development with commercial sustainability.

CONCLUSION

The consolidated development of the Omnivest project demonstrates a complete and coherent evolution from concept to scalable, AI-enabled, and market-ready platform. Across fifteen sprints, the project successfully integrated user-centered design, secure full-stack engineering, cloud infrastructure, advanced predictive intelligence, and business validation. This consolidated report highlights the robustness of the technical architecture, the maturity of the data and AI pipeline, and the strategic alignment with real-world financial and operational goals, positioning Omnivest for future production deployment and growth.

REFERENCES

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